

3.3 Characteristics of the tropical rainforest ecosystem

Tropical rainforests are found between the Tropic of Cancer (23.5°N) and the Tropic of Capricorn (23.5°S), mainly within 10° to 15° of the equator. The three main areas of tropical rainforest are in the islands of South-East Asia, the Congo Valley of Africa, and the Amazon Valley of South America. Tropical rainforests extend up to approximately 1000 metres above sea level. They are not found above 1000 metres as the cooler temperatures do not support the tropical rainforests vegetation type. The equatorial climate is hot, wet, and humid all year, with a low **diurnal** temperature range. This is due to a combination of factors that include: the high rate of sunlight (insolation) all year-round as the sun remains almost directly over the equator (0° latitude) for most of the year, and the presence of a low-pressure system due to the year-round heating of the Earth by the sun. Table 3.3 summarises these factors and the resulting climatic characteristics.

Table 3.3: The climatic characteristics of the tropical rainforest ecosystem

Climate characteristic	Impact on climate
Latitude	Distance from the equator impacts how much sunlight a region receives annually. The equatorial zone (the equator and up to 10°–15° north and south of the equator) receives high rates of sunlight all year-round.
High rates of insolation	Year-round heating of the Earth's surface due to high rates of sunlight results in higher temperatures and the development of a low-pressure system.
Low pressure system	Warm and moist air rises in the atmosphere and cools to form clouds that bring regular precipitation.
Climate factor	Climatic factor range
Temperature	Average daily temperatures between 25 °C and 28 °C
Humidity	Averages 90% in wet season and 70% in dry season
Rainfall	Average annual rainfall between 2000 mm and 10 000 mm
Diurnal temperature range	1 °C to 3 °C

Tropical rainforests are divided vertically into four distinctive layers (Source A). The layers, starting from the ground, are the rainforest floor, the understorey, the canopy, and the emergent layer. Different species of tall trees are the dominant vegetation type in tropical rainforests. Some trees exceed 30 metres in height. Some of the trees have wide buttress roots that surround the base and help to support their size. The trees are often smooth-barked which helps the trees prevent vines and epiphytes growing up their trunks. They are also evergreen which supports the high rates of photosynthesis all year around. Other plants in tropical rainforests include plants that are able to live in the humid, low level of light found below the larger trees, for example, herbaceous plants and tree ferns. Other plants, such as vines, lianas, and epiphytes, have adapted by living on other trees and growing on or up their trunks to seek the light. The vegetation on the ground layer is often dominated by herbaceous plants that can grow in lower levels of light. Tropical rainforests include wetland, mangrove, and river ecosystems where soils can be permanently or seasonally saturated.



Figure 3.1: A tropical rainforest tree showing well-developed buttress roots



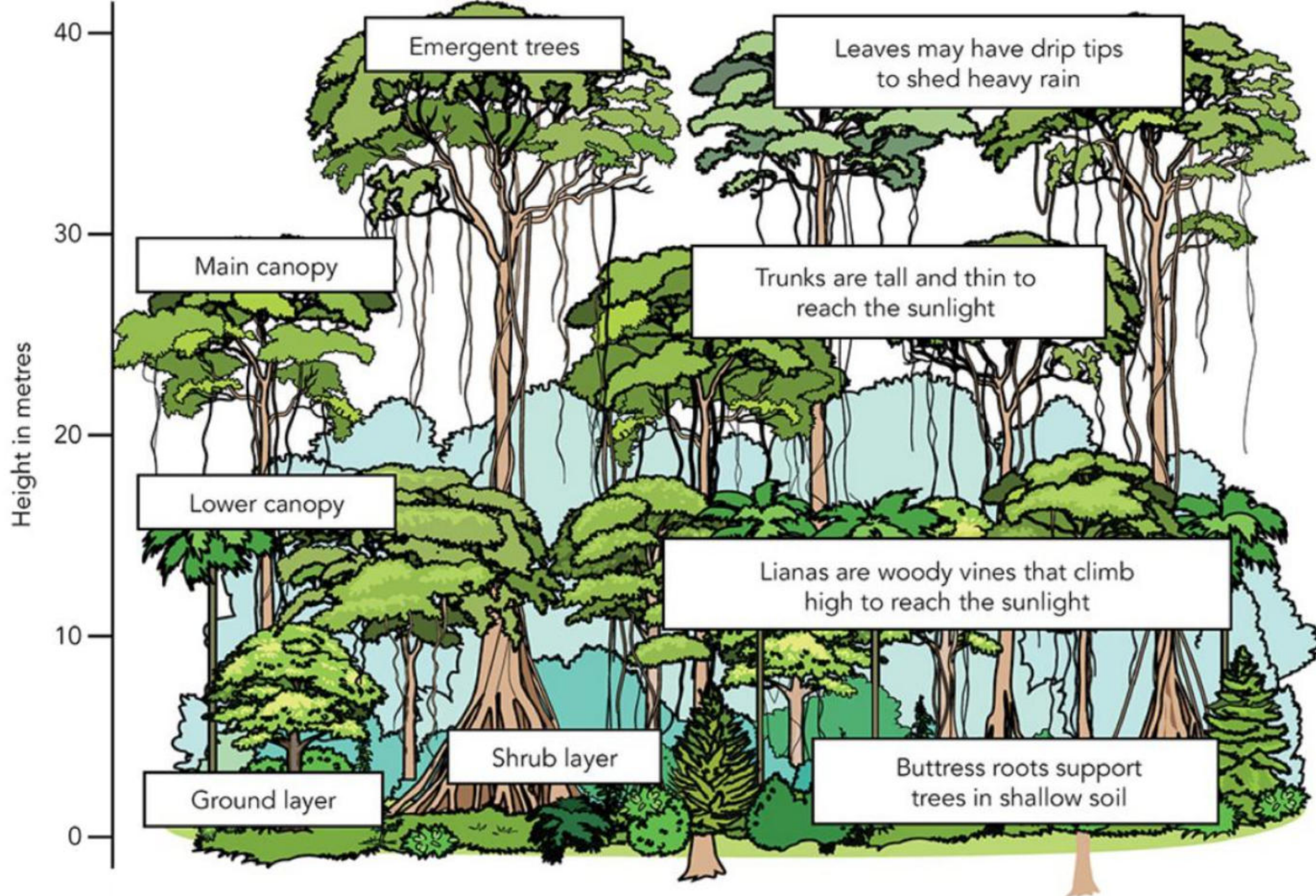
Figure 3.2: Lianas growing up the trunk of a tree



Figure 3.3: Epiphytes growing on the trunk of a tree



Figure 3.4: A tree fern growing on the side of a valley



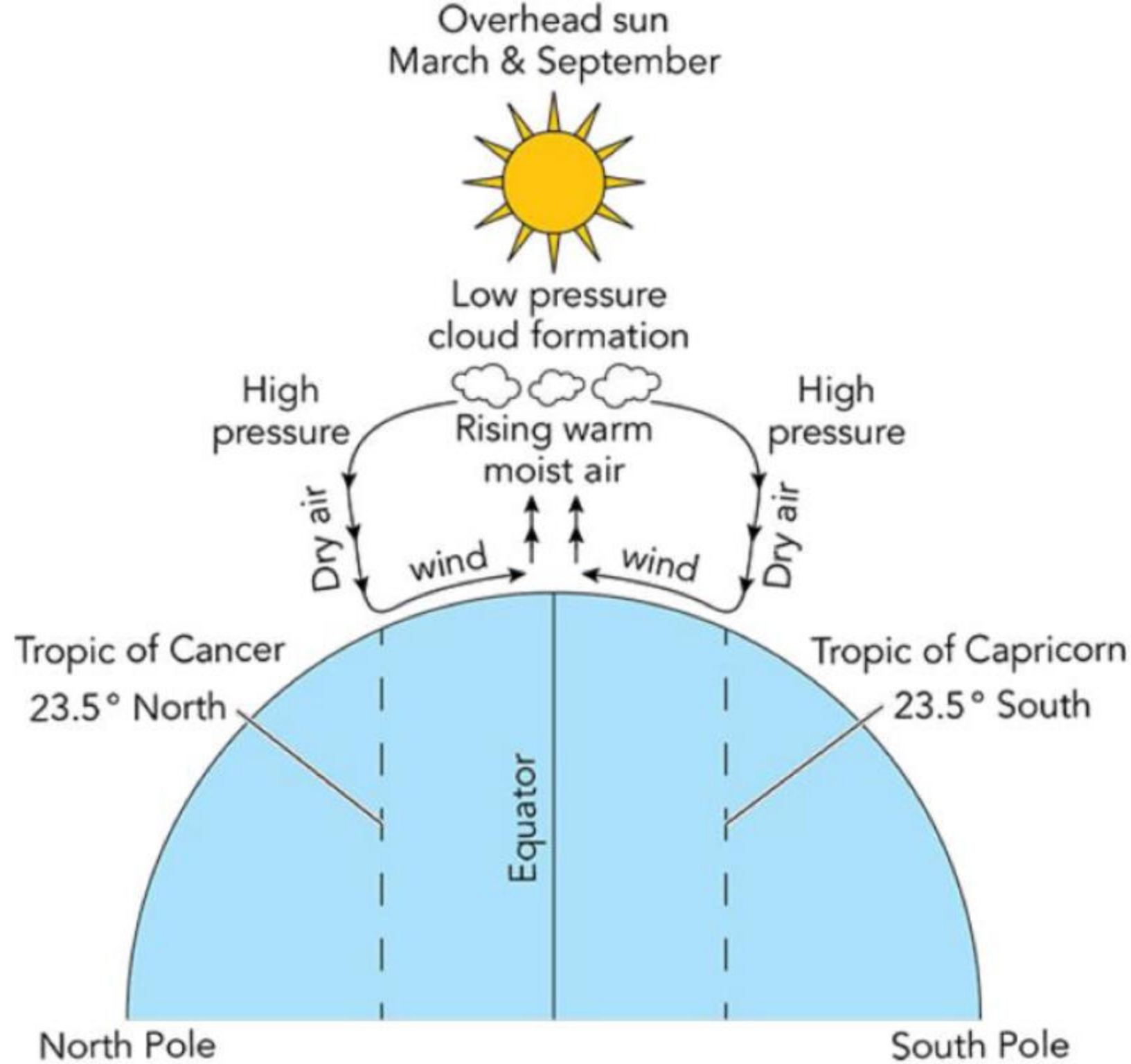
Source **A**: Vertical layers within the tropical rainforest

TASK 1: Study Source **A**

- Describe the structure of a tropical rainforest.
- Why does so little light reach the rainforest floor? How does that impact the plants on the rainforest floor?
- Explain how a reduction in sunlight would change this ecosystem.

Abiotic and biotic factors of tropical rainforest ecosystems

The key abiotic factor that influences the development of tropical rainforests in these regions is the high rate of sunlight year-round. The sun is directly over the equator (0° latitude) for much of the year. This intense continuous sunlight results in warm temperatures that cause high rates of evaporation. The warm, moist air rises, forming a low-pressure system. As the air rises, it cools. This causes the water vapour to **condense** and form clouds, which then cause daily **convective rainfall**.

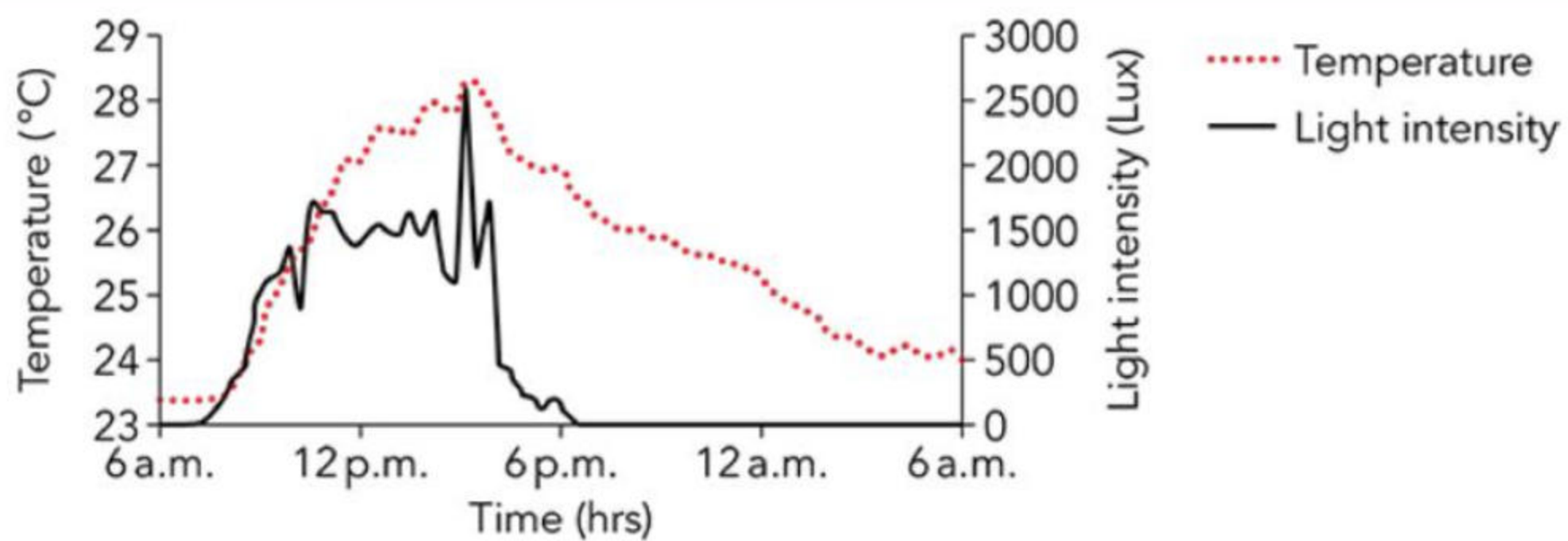


Source B: Diagram showing why the equatorial region has high levels of rainfall all year around

TASK 2: Study Source B

This diagram shows the low-pressure system and formation of clouds over the equator (0° latitude) when the sun is directly overhead. In June, the sun is over the Tropic of Cancer, and in December, it is over the Tropic of Capricorn.

- Explain why clouds and thunderstorms form over the equator.
- Explain why storms take place mostly later in the afternoon. You can also use Source C to help you answer this question.
- Explain why the equatorial region does not have summer and winter seasons.



Source C: Temperature and light intensity over a 24-hour period in a tropical rainforest

TIP

Warm, rising air forms a low-pressure zone, while cold, sinking air forms a high-pressure zone. Be careful not to mix these up. Warm, moist air is less dense and rises (low-pressure). Cold, dry air is dense and sinks (high-pressure).

TASK 3: Study Source C

- Using the highest temperature as 28.3°C and the lowest temperature as 23.4°C , **calculate** the daily temperature range for the tropical rainforest.
- With reference to Source C, **compare the change** in light intensity with the change in temperature in a tropical rainforest over a 24-hour period.

c Explain the relationship between the light intensity and temperature in the rainforest.

The high rates of daily light and high temperatures support high rates of photosynthesis. This directly influences the biotic characteristics of the tropical rainforest. Vegetation can grow rapidly. This rapid growth causes plants to compete for light, growing taller than the plants around them to reach for the light above. On the rainforest floor, light is limited as the tree canopy blocks light to the lower levels of the rainforest. Some plants, like vines and epiphytes, grow on and up the trees to support them as they reach for the light.

Animals use the vegetation for shelter and as hiding places from predators. With such diverse habitat, there are many different animals competing for food. Many have adapted to eat particular food that is not eaten by other animals. Many of the animals never touch the ground, living their entire lives up in the trees. The plants also depend on the animals for survival, as the animals pollinate flowers and distribute seeds. This is a crucial factor in supporting plant reproduction.



Source D: A long-tailed hermit hummingbird found in the Amazon rainforest

TASK 4: Study Source D

- a** In pairs, discuss how you think the long-tailed hermit hummingbird has adapted for feeding from flowers found in the Amazon rainforest. Suggest three features that show how it has specialised for this type of feeding.
- b** In pairs, research a rainforest animal, bird, or insect that has adapted to live in the tropical rainforest. **Create a diagram**, such as a mind map or flowchart, giving the information on the organism you have identified. Share your findings with the rest of the class.

REFLECTION

Does creating a diagram of the adaptations of a rainforest animal help you understand the causes for their adaptations? Are you able to apply that thinking to other animals you know of that live in the rainforest?